

Track 53: Fashion Technology

June 2026

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Tier 3 — Specialized | Department: Fashion Technology

Skill path: *Digital Illustration & Styling → Technical Flat Sketching & Tech Packs → Textile & Print Design → Pattern Making & Construction → 3D Virtual Draping (CLO3D) → Merchandising → Sustainable Fashion*

Related: [Track 09 — UI/UX & Web Design](#) · [Track 11 — Digital Marketing](#) · [Department Map](#)

Teachable Skill Modules

Industry-ready skills you can deliver to students, grouped by module. Each skill includes suggested tools and the earliest tier where it is taught.

Module A — Digital Design & Visual Communication

Skill	Tools / Software	Introduced
Digital Fashion Illustration (figure drawing, rendering, fabric texture)	Adobe Illustrator, Procreate, Photoshop, Krita	2 Weeks
Digital Mood Boarding & Conceptual Visual Styling (theme, color story, trend direction)	Canva, Pinterest, Milanote, Adobe Express, Figma	2 Weeks
Lookbook & Campaign Layout Design	InDesign, Canva, Adobe Express	1 Month
Fashion Photography Direction & Styling Basics (on-model vs flat lay)	Smartphone + Lightroom; studio lighting intro	3 Months
Brand Identity for Fashion Labels (logo, label, hang tag, packaging mockups)	Illustrator, Canva; links to Track 09	6 Months

Module B — Technical Design & Production Documentation

Skill	Tools / Software	Introduced
Technical Flat Sketching & CAD Drafting (front/back views, stitch lines, callouts)	Adobe Illustrator, CorelDRAW, Optitex (2D)	1 Month
Digital Tech Pack Creation & Specification Sheet Design (BOM, measurements, construction notes)	Excel/Google Sheets + Illustrator; Techpacker, PLM intro	3 Months
Grading & Size Set Documentation	Manual grading + Gerber Accumark intro	3 Months
Costing Sheet & Bill of Materials (BOM) Preparation	Excel, Tally basics (Track 10)	3 Months
Production File Handoff for Sampling & Bulk Manufacturing	PDF tech packs, marker planning intro	6 Months

Module C — 3D Virtual Fashion & Digital Sampling

Skill	Tools / Software	Introduced
3D Virtual Garment Draping & Avatar Customization	CLO 3D, Browzwear (intro), Marvelous Designer (intro)	3 Months
Digital Fabric Simulation & Trim Application	CLO 3D material library, custom fabric scans	3 Months
Virtual Runway / Lookbook Renders (turntable, animation export)	CLO 3D, Blender (optional), Photoshop compositing	6 Months
Digital Sample Approval Workflow (reduce physical sampling cycles)	CLO 3D + tech pack integration	6 Months

Module D — Textile, Print & Surface Design

Skill	Tools / Software	Introduced
Textile Pattern, Repeat & Print Design (placement vs all-over repeat)	Illustrator, Photoshop, Procreate, Reppper	1 Month
Color Separation & Print-Ready File Export	Illustrator, Photoshop	3 Months
Fabric Swatch Documentation & Material Specification	Physical swatches + digital spec sheets	2 Weeks
Sustainable Material Selection & Eco-Label Awareness	GOTS, OEKO-TEX, BCI reference frameworks	6 Months
Zero-Waste & Upcycling Design Techniques	Manual pattern manipulation + CLO 3D simulation	6 Months

Module E — Pattern Making, Draping & Garment Construction

Skill	Tools / Software	Introduced
Flat Pattern Drafting (bodice, skirt, sleeve, collar, trouser blocks)	Pattern paper, grading rulers, dress forms	2 Weeks
Garment Draping on Dress Form	Muslin, pins, dress forms	1 Month
Toile Fitting & Pattern Correction	Sewing lab, fitting sessions	1 Month
Industrial Sewing & Finishing Techniques	Sewing machines, overlock, pressing	1 Month
CAD Pattern Making & Marker Making Intro	Gerber Accumark, Optitex, Lectra (intro)	6 Months

Module F — Fashion Business, Merchandising & Marketing

Skill	Tools / Software	Introduced
Range Planning & Seasonal Merchandising	Excel, merchandise calendar templates	6 Months
Trend Forecasting & Market Research	WGSN (intro), Pinterest Trends, runway analysis	1 Month
Fashion E-Commerce & Social Commerce Basics	Shopify, Instagram Shop; Track 11	6 Months
Export Documentation & Apparel Production Management	Invoice, packing list, HS code awareness	9 Months
Collection Pitch & Buyer Presentation	PowerPoint/Canva pitch deck + physical/digital portfolio	6 Months

Skill Progression by Duration

Duration	Core skills to teach
2 Weeks	Digital Fashion Illustration · Digital Mood Boarding · Fabric Swatch Documentation · Basic Flat Pattern Drafting
1 Month	+ Technical Flat Sketching · Textile Pattern & Print Design · Garment Draping · Toile Fitting · Trend Forecasting
3 Months	+ Digital Tech Pack Creation · 3D Virtual Garment Draping (CLO3D) · Color Separation · Grading Documentation · Lookbook Design
6 Months	+ Virtual Runway Renders · CAD Pattern Making Intro · Range Planning · Sustainable Design · Brand Identity · E-Commerce Basics
9 Months	+ Production Handoff · Export Documentation · Graduate Collection · Brand Launch Pitch · Fashion Show / Campaign

Track Overview

Who this is for: Fashion design students, textile graduates, aspiring fashion designers, merchandisers, and entrepreneurs launching apparel brands.

Prerequisites: Basic drawing skills helpful; no prior fashion education required for 2-week/1-month tiers.

Tools: Adobe Illustrator · Photoshop · Procreate · CLO 3D · Canva · InDesign · Techpacker (intro) · Gerber Accumark (intro) · sewing lab equipment · dress forms · fabric swatches

Certification targets: Portfolio-ready for design houses; Adobe Certified Professional (optional, 6-month)

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Fashion awareness	Illustration board + basic garment
1 Month	80–100	Junior design assistant	Mini collection (3 looks)
3 Months	250–300	Internship-ready	Capsule collection + lookbook
6 Months	500–600	Job-ready designer/technologist	Portfolio + fashion CAD project
9 Months	750–900	Industry-ready	Graduate collection + brand pitch

Weekly Rhythm (All Durations)

Mon–Thu design/construction labs | Fri project milestone | Sat critique session | Sun portfolio documentation

2-Week Crash Course

Mini Project: Fashion illustration board + stitched sample | Level: L1

Days	Topics	Deliverable	Self-Check
D1–2	Fashion history, elements of design, color theory; Digital Mood Boarding & Conceptual Visual Styling	Mood board (Canva/Milanote)	Name 5 elements of design
D3–4	Digital Fashion Illustration — croquis, rendering (Illustrator/Procreate)	5 illustrated figures	Render 1 fabric texture
D5–6	Textile basics: fibers, weaves, finishes; Fabric Swatch Documentation	Fabric swatch book (10 samples)	Identify 3 weave types
D7–8	Flat Pattern Drafting intro: bodice block, skirt block	Paper patterns	Draft 1 block correctly
D9–10	Basic garment construction, capstone	Illustration board + stitched sample	

Submission checklist: - [] Mood board with theme statement - [] 5 fashion illustrations with color - [] Fabric swatch book labeled - [] One completed stitched garment or sample

1-Month Foundation

Capstone: Mini collection (3 coordinated looks) | Level: L2

Week	Topics	Project
W1	Digital Fashion Illustration advanced, Digital Mood Boarding , trend research	Trend report + illustrations
W2	Textile science, Textile Pattern, Repeat & Print Design (Illustrator/Procreate)	Print design + fabric specification sheet
W3	Technical Flat Sketching , pattern drafting, toile fitting	Flat sketches + pattern set for 1 design
W4	Garment construction, Lookbook Layout Design , finishing	Mini collection (3 looks) + lookbook PDF

W4 acceptance criteria: - [] 3 coordinated designs with illustrations - [] At least 1 garment fully constructed - [] Lookbook PDF (6+ pages) - [] Costing sheet (basic)

3-Month Job Ready

Capstone: Capsule collection + fashion CAD | **Level:** L3

Month	Topics	Projects
M1	Advanced illustration, Conceptual Visual Styling , collection planning	Design portfolio (10+ sketches)
M2	Pattern making, draping, Technical Flat Sketching & CAD Drafting	Pattern library + flat sketch set
M3	3D Virtual Garment Draping & Avatar Customization (CLO3D) , Digital Tech Pack Creation & Specification Sheets , capstone	Capsule collection (5 looks) + tech packs + CLO3D renders

M3 Capstone acceptance criteria: - [] 5-look capsule with cohesive theme and mood board - [] **Digital tech packs** for each look (flat sketches, BOM, measurements, construction notes) - [] **CLO3D virtual garment** with avatar customization OR fully constructed hero piece - [] **Textile print/pattern design** applied to at least 1 look - [] Lookbook + cost sheet

6-Month Professional

Month	Topics	Projects
M1	Digital illustration, mood boarding, textiles, print design	Trend-driven design portfolio
M2	Pattern making, draping, garment construction	Ready-to-wear garment line (3 pieces)
M3	CLO3D virtual draping , digital fabric simulation, tech pack production workflow	Digital collection + approved tech packs
M4	Merchandising, range planning, costing, sourcing	Merchandise plan for season
M5	Sustainable fashion, zero-waste design, brand identity for fashion labels	Sustainable design project + brand kit
M6	Virtual runway renders , portfolio development, fashion show prep, capstone	Industry-ready portfolio + brand pitch

M6 Capstone acceptance criteria (L4): - [] Portfolio with 3+ completed projects - [] Range plan

with costing and margin analysis - [] Digital + physical sample of hero garment - [] Brand identity (logo, label, lookbook) - [] 10-minute collection presentation

9-Month Advanced

Capstones: Graduate fashion collection + fashion brand launch plan | **Level:** L5

Additional modules: - Couture techniques intro - Fashion marketing ([Track 11](#) integration) - E-commerce for fashion brands - Internship with boutique/design house (optional) - Fashion photography and styling - Export documentation and apparel production management

Phase	Focus
M1–M6	Complete 6-month curriculum
M7	Advanced draping + couture finishing techniques
M8	Production planning, factory liaison, quality control
M9	Graduate collection (8+ looks) + brand launch pitch + fashion show

Capstone 1 — Graduate Collection: - [] 8+ look cohesive collection - [] Full tech pack package - [] Runway or lookbook photoshoot - [] Production cost analysis

Capstone 2 — Brand Launch Plan: - [] Brand identity system - [] E-commerce or social commerce strategy - [] 3-season merchandise calendar - [] Investor/customer pitch deck

Assessment Rubric (Track-Specific)

Component	Weight	Criteria
Weekly Quiz	20%	Textiles, design principles, terminology
Lab Completion	40%	Illustrations, patterns, construction quality
Project	30%	Collection cohesion, craftsmanship, presentation
Portfolio	10%	Lookbook, tech packs, digital portfolio

Recommended Resources

- Adobe Illustrator & Procreate fashion illustration tutorials
 - [CLO 3D learning resources](#) — virtual draping, avatar customization, animation export
 - Techpacker / WGSN (intro) for tech pack and trend reference
 - Textile science reference: *Fabric for Fashion* (Clive Hallett)
 - [Track 09 — UI/UX](#) for brand website; [Track 11 — Digital Marketing](#) for fashion campaigns
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Cross-Track Integration

Need	Track
Visual design & branding	09 UI/UX & Web Design
Fashion brand marketing	11 Digital Marketing
E-commerce website	01 Full Stack Development
Business documentation	10 Office Suite & Tally
Career & portfolio prep	14 Career Development
College fashion project support	12 School & College Projects

Appendix: Mentor Notes

Lab requirements: Sewing machines, dress forms, ironing stations, cutting tables, minimum 2m fabric per student per project.

Common pitfalls: Skipping toile fittings; poor fabric grain alignment; illustrations without proportion; tech packs missing measurements.

Optional stretch goals: Participate in local fashion show; collaborate with [Track 11](#) for brand launch campaign.

Recommended combo: Fashion Technology + Digital Marketing (6-month) for designer-entrepreneurs.

Related Documents

- [Tier Demand Catalog](#)
- [Department Course Map](#)
- [Combo Packages](#)